Porter-Thomas Statistics under Two Sampling Regimes

Why "many shots, one circuit" and "many circuits, one shot"
give the same expected F_{XEB} but different concentration

A side-by-side variance analysis

Quip Protocol Project — Internal Reference Document

June 2026

Abstract

Both the Aaronson–Hung (AH) and Liu et al. protocols sample from random quantum circuits and use the cross-entropy benchmark F_{XEB} as their acceptance test. They differ in *how* the samples are collected: AH uses one circuit with m shots per round; Liu uses m distinct circuits with one shot each. This document establishes that both regimes yield identically Porter-Thomas-distributed observed probabilities, produce the same expected F_{XEB}, but differ in their variance structure in a way that has direct consequences for certified-entropy bounds. The core result is a variance decomposition (§5) demonstrating that AH's samples are correlated through the shared circuit while Liu's samples are independent across rounds. This correlation gap explains why AH requires the Entropy Accumulation Theorem for its concentration argument and gives a looser bound, while Liu's i.i.d. structure admits direct Chernoff concentration and a tighter certified-entropy claim for the same hardware. Quip's v0.4 implementation adopts Liu's structure.

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1 Setup and Notation

We consider an n-qubit quantum circuit C drawn from a distribution sufficiently close to the Haar measure on $U(2^n)$. Throughout this document, write $N := 2^n$ for the dimension of the computational basis. For any circuit C and bitstring $x \in \{0,1\}^n$, define

$$p_C(x) := |\langle x \mid U_C \mid 0^n\rangle|^2,$$

the probability that an ideal noiseless execution of C produces measurement outcome x.

A real (noisy) quantum sampler with fidelity $\phi \in [0,1]$ is modelled as producing x from the mixture distribution

$$q_C(x) = \phi \cdot p_C(x) + (1-\phi) \cdot \frac{1}{N},$$

i.e., the ideal distribution with probability ϕ and uniform noise with probability $1-\phi$.

1.1 The two sampling regimes

We distinguish:

Regime A (Aaronson–Hung): Pick one circuit C. Take m samples $x_1, x_2, \ldots, x_m$ i.i.d. from q_C. The m samples share the same underlying circuit.

Regime B (Liu et al.): Pick m independent circuits $C_1, C_2, \ldots, C_m$. From each C_i, take exactly one sample $x_i \sim q_{C_i}$. Each (C_i, x_i) pair is independent of every other.

The cross-entropy benchmarking score is identical in form:

$$F_{\text{XEB}} := \frac{N}{m} \sum_{i=1}^{m} p_{C_i}(x_i) - 1, \tag{1}$$

where in Regime A, all $C_i = C$ (the shared circuit), and in Regime B each C_i is independent. The protocol accepts if $F_{\text{XEB}} \geq \chi$ for some threshold χ (Liu uses $\chi = 0.3$).

1.2 What we want to establish

Two questions:

(1) Is the *statistical distribution* of $p_{C_i}(x_i)$ identical in the two regimes? **Yes.**

(2) Is the *concentration behaviour* of F_{XEB} identical in the two regimes? **No.**

The first answer is the source of common confusion — both regimes produce Porter-Thomas statistics, so it's tempting to conclude they're interchangeable. The second answer is why they're not, and why Liu's structure is preferred in v0.4.

2 The Porter-Thomas distribution

We collect the relevant facts about $p_C(x)$ for Haar-random circuits.

Theorem 2.1 (Porter–Thomas)**.** *Let C be a circuit whose unitary is Haar-distributed on $U(N)$. For any fixed bitstring $x \in \{0,1\}^n$, the random variable $N \cdot p_C(x)$ converges in distribution to* $\mathrm{Exp}(1)$ *as $N \to \infty$.*

The proof appears in Porter and Thomas (1956) and is reviewed in Boixo et al. (2018). Sufficiently-deep random circuits approximate the Haar distribution closely enough that the convergence holds in practice once $d \gtrsim n$ (where d is the circuit depth).

2.1 Moments of Porter-Thomas

For $Y \sim \mathrm{Exp}(1)$, the moment generating function is $\mathbb{E}[e^{tY}] = 1/(1-t)$ for $t < 1$, giving moments $\mathbb{E}[Y^k] = k!$. Since $N \cdot p_C(x) \sim Y$, we have

$$\mathbb{E}[p_C(x)^k] = \frac{k!}{N^k}. \tag{2}$$

The first three moments will appear throughout:

$$\mathbb{E}[p_C(x)] = \frac{1}{N}, \qquad \mathbb{E}[p_C(x)^2] = \frac{2}{N^2}, \qquad \mathbb{E}[p_C(x)^3] = \frac{6}{N^3}.$$

2.2 The collision probability identity

A computation we'll use:

Lemma 2.2. *For Haar-random C,*

$$\mathbb{E}_C\left[\sum_{x\in\{0,1\}^n} p_C(x)^2\right] = \frac{2}{N}.$$

Proof. By linearity and (2), $\mathbb{E}_C[\sum_x p_C(x)^2] = N \cdot \mathbb{E}_C[p_C(x)^2] = N \cdot 2/N^2 = 2/N$. □

This identity is what produces the famous "factor of 2" in $\mathrm{F}_{\mathrm{XEB}}$ calculations.

3 The size-biased distribution: $p_C(x)$ when $x \sim p_C$

Both regimes involve a quantity of the form $p_{C_i}(x_i)$ where x_i *was actually drawn from* q_{C_i} (not a fixed bitstring chosen independently of the circuit). This conditional choice changes the distribution.

Intuition

If you sample from a distribution, you're more likely to observe bitstrings the distribution assigned high probability to. So the values of $p_C(x)$ you see when $x \sim p_C$ are biased toward larger values than $p_C(x)$ for a fixed independent x. This is called *size-biased sampling.*

3.1 Conditional distribution: ideal sampler

For the noiseless case $\phi = 1$ (so $q_C = p_C$):

Proposition 3.1. *Let C be Haar-random and $X \sim p_C$. Then for any measurable bounded f,*

$$\mathbb{E}[f(p_C(X))] = N \cdot \mathbb{E}[p_C(x_0)\, f(p_C(x_0))]$$

where x_0 is any fixed bitstring. In particular:

$$\begin{aligned}
\mathbb{E}[p_C(X)] &= \frac{2}{N}, \\
\mathbb{E}[p_C(X)^2] &= \frac{6}{N^2}, \\
\mathrm{Var}[p_C(X)] &= \frac{6}{N^2} - \frac{4}{N^2} = \frac{2}{N^2}.
\end{aligned}$$

Proof. For any function f of the probability,

$$\begin{aligned}
\mathbb{E}_{C,X\sim p_C}[f(p_C(X))] &= \mathbb{E}_C\left[\sum_x p_C(x) f(p_C(x))\right] \\
&= N \cdot \mathbb{E}_C[p_C(x_0) f(p_C(x_0))]
\end{aligned}$$

where the second line uses symmetry over x (Haar-randomness makes all bitstrings statistically equivalent). Setting $f(y) = y^k$ and applying (2) gives

$$\mathbb{E}[p_C(X)^k] = N \cdot \mathbb{E}[p_C(x_0)^{k+1}] = N \cdot \frac{(k+1)!}{N^{k+1}} = \frac{(k+1)!}{N^k}.$$

Setting $k = 1, 2$ gives the stated values. □

Intuition

The conditional distribution of $N \cdot p_C(X)$ when $X \sim p_C$ is the *Gamma*$(2, 1)$ distribution: a sum of two independent Exp(1)'s, with mean 2. Compare to $N \cdot p_C(x_0)$ for fixed x_0, which is Exp(1) with mean 1. The size-biased version is literally "one extra exponential's worth" of probability mass.

3.2 Conditional distribution: noisy sampler

For $\phi < 1$, $X \sim q_C = \phi p_C + (1-\phi)/N$, so by linearity:

$$\mathbb{E}[p_C(X) \mid C] = \phi \cdot \mathbb{E}_{X\sim p_C}[p_C(X) \mid C] + (1-\phi) \cdot \mathbb{E}_{X\sim U}[p_C(X) \mid C]. \tag{3}$$

The first term gives ϕ times the size-biased mean; the second term gives $(1-\phi)/N$. Averaging over C:

$$\mathbb{E}[p_C(X)] = \phi \cdot \frac{2}{N} + (1-\phi) \cdot \frac{1}{N} = \frac{1+\phi}{N}. \tag{4}$$

This is the key identity: **the expected probability of an observed sample under a fidelity-ϕ sampler is $(1 + \phi)/N$.**

4 Expected $\mathbf{F_{XEB}}$ — identical in both regimes

We can now compute $\mathbb{E}[F_{\mathrm{XEB}}]$ in both regimes.

4.1 Regime B (Liu)

In Regime B, each (C_i, x_i) is an independent draw. Using (4):

$$\mathbb{E}[F_{\mathrm{XEB}}^{(B)}] = \frac{N}{m}\sum_{i=1}^{m}\mathbb{E}[p_{C_i}(x_i)] - 1 = N \cdot \frac{1+\phi}{N} - 1 = \phi. \tag{5}$$

4.2 Regime A (AH)

In Regime A, all m samples come from the same C. We need the law of total expectation:

$$\begin{aligned}
\mathbb{E}[F_{\mathrm{XEB}}^{(A)}] &= \mathbb{E}_C\left[\frac{N}{m}\sum_{i=1}^{m}\mathbb{E}[p_C(x_i) \mid C] - 1\right] \\
&= \mathbb{E}_C\left[\frac{N}{m} \cdot m \cdot \mathbb{E}[p_C(x_1) \mid C] - 1\right] \\
&= N \cdot \mathbb{E}_C[\mathbb{E}[p_C(x_1) \mid C]] - 1 \\
&= N \cdot \mathbb{E}[p_C(x_1)] - 1 \\
&= N \cdot \frac{1+\phi}{N} - 1 = \phi.
\end{aligned} \tag{6}$$

Intuition

Both regimes give $\mathbb{E}[F_{\mathrm{XEB}}] = \phi$. This is what makes $\mathrm{F_{XEB}}$ a useful fidelity estimator in both settings. The *mean* is structure-agnostic — it depends only on the per-sample expectation, which involves Porter-Thomas size-biasing in the same way for both regimes.

Remark. For an honest quantum sampler with $\phi \approx 1$, $\mathbb{E}[F_{\mathrm{XEB}}] \approx 1$. For a uniform-random classical adversary with $\phi = 0$, $\mathbb{E}[F_{\mathrm{XEB}}] = 0$. The Liu protocol's threshold $\chi = 0.3$ sits between these, corresponding to roughly the fidelity Quantinuum H2 delivers on the 56-qubit circuit ensemble.

5 Variance — where the regimes differ

Now we come to the central calculation. We compute $\mathrm{Var}[F_{\mathrm{XEB}}]$ in each regime.

5.1 Variance in Regime B (Liu)

Each term $p_{C_i}(x_i)$ in the sum is an independent random variable (independent across i because the (C_i, x_i) pairs are independent). So

$$\mathrm{Var}[F_{\mathrm{XEB}}^{(B)}] = \frac{N^2}{m^2} \cdot \sum_{i=1}^{m}\mathrm{Var}[p_{C_i}(x_i)] = \frac{N^2}{m} \cdot \mathrm{Var}[p_C(X)], \tag{7}$$

where in the last step we use that the m variances are equal.

We computed $\mathrm{Var}[p_C(X)] = 2/N^2$ for the noiseless case (Proposition 3.1). For $\phi < 1$ the variance changes by an $O(1)$ factor; what matters is that

$$\mathrm{Var}[F_{\mathrm{XEB}}^{(B)}] = \frac{c(\phi)}{m} \tag{8}$$

for some constant $c(\phi) = O(1)$. **Variance scales as $1/m$.**

5.2 Variance in Regime A (AH)

For Regime A we use the *law of total variance*:

$$\mathrm{Var}[F_{\mathrm{XEB}}^{(A)}] = \mathbb{E}_C\big[\mathrm{Var}[F_{\mathrm{XEB}}^{(A)} \mid C]\big] + \mathrm{Var}_C\big[\mathbb{E}[F_{\mathrm{XEB}}^{(A)} \mid C]\big]. \tag{9}$$

We compute each term.

First term: within-circuit variance

Given C fixed, the samples $x_1, \ldots, x_m$ are i.i.d. from q_C. So

$$\mathrm{Var}[F_{\mathrm{XEB}}^{(A)} \mid C] = \frac{N^2}{m^2} \cdot m \cdot \mathrm{Var}[p_C(X) \mid C] = \frac{N^2}{m} \cdot \mathrm{Var}[p_C(X) \mid C]. \tag{10}$$

Taking the outer expectation:

$$\mathbb{E}_C\big[\mathrm{Var}[F_{\mathrm{XEB}}^{(A)} \mid C]\big] = \frac{N^2}{m} \cdot \mathbb{E}_C[\mathrm{Var}[p_C(X) \mid C]]. \tag{11}$$

This is the term that scales as $1/m$. So far, indistinguishable from Regime B.

Second term: between-circuit variance — the new term

The conditional expectation given C is $\mathbb{E}[F_{\mathrm{XEB}}^{(A)} \mid C] = N \cdot \mathbb{E}[p_C(X) \mid C] - 1$. This is itself a random variable (depending on which C was drawn).

For the noiseless case, $\mathbb{E}[p_C(X) \mid C] = \sum_x p_C(x)^2$ (this is the *collision probability* of distribution p_C). By Lemma 2.2, its expectation is $2/N$. Its variance is non-trivial:

Lemma 5.1. *For Haar-random C,*

$$\mathrm{Var}_C\left[\sum_x p_C(x)^2\right] = \frac{20}{N^3} + O(N^{-4}).$$

(The leading-order computation involves $\mathbb{E}_C[(\sum_x p_C(x)^2)^2] = 4/N^2 + 20/N^3 + O(N^{-4})$, which follows from Weingarten calculus on the Haar measure. We don't reproduce the calculation here; see the Boixo et al. supplement.)

So:

$$\begin{aligned}
\mathrm{Var}_C\big[\mathbb{E}[F_{\mathrm{XEB}}^{(A)} \mid C]\big] &= \mathrm{Var}_C\big[N \cdot \mathbb{E}[p_C(X) \mid C] - 1\big] \\
&= N^2 \cdot \mathrm{Var}_C[\mathbb{E}[p_C(X) \mid C]] \\
&\approx N^2 \cdot \frac{20}{N^3} \\
&= \frac{20}{N}.
\end{aligned} \tag{12}$$

Crucially, this term does not depend on m.

Putting it together

Combining (11) and (12):

$$\mathrm{Var}[F_{\mathrm{XEB}}^{(A)}] = \underbrace{\frac{c(\phi)}{m}}_{\text{within-circuit}} + \underbrace{\frac{20}{N} + O(\cdot)}_{\text{between-circuit}} . \tag{13}$$

5.3 Comparison

Compare:

$$\begin{aligned}\mathrm{Var}[F_{\mathrm{XEB}}^{(B)}] &= \frac{c(\phi)}{m},\\ \mathrm{Var}[F_{\mathrm{XEB}}^{(A)}] &= \frac{c(\phi)}{m} + \frac{20}{N} + O(\cdot).\end{aligned}$$

For $n = 56$, $N = 2^{56} \approx 7.2 \times 10^{16}$, so $20/N \approx 3 \times 10^{-16}$. For $m = 1522$, $c(\phi)/m \sim 1/m \approx 6.6 \times 10^{-4}$. So the between-circuit term is dwarfed by the within-circuit term.

Does this mean the difference is negligible? No — and this is the subtle point. The variance values *evaluated for a single random circuit* are similar in magnitude. But the structural difference manifests when we ask about *concentration over many independent protocol executions*.

Intuition

The two regimes give similar single-run variance because the between-circuit fluctuation is small (N is huge, so individual circuits all look approximately Porter-Thomas). The structural difference is not in the magnitude of variance but in **which concentration inequality applies**: i.i.d. samples (Regime B) admit standard Chernoff bounds; correlated samples sharing a circuit (Regime A) require the Entropy Accumulation Theorem.

6 Concentration: where the structural difference bites

The certified-entropy bound depends not just on the variance of $\mathrm{F}_{\mathrm{XEB}}$ but on *tail bounds* — the probability that $\mathrm{F}_{\mathrm{XEB}}$ deviates from its expectation by more than δ, for various δ.

6.1 Regime B: direct Chernoff

In Regime B, the m summands $p_{C_i}(x_i)$ are i.i.d. bounded random variables (bounded by 1, since they're probabilities). The Chernoff bound (or Hoeffding's inequality) gives

$$\mathbb{P}[|F_{\mathrm{XEB}}^{(B)} - \phi| > \delta] \leq 2\exp(-c \cdot m\delta^2), \tag{14}$$

for some constant c. The protocol's soundness analysis uses exactly this bound (with a careful split of the soundness budget $\varepsilon_{\mathrm{sou}}/2$ between this Chernoff term and the hypergeometric tail; see Liu et al. Lemma 2 and our own implementation in `leg5_qmin_liu.compute_q_min`).

Crucially: nothing more sophisticated is needed. The bound is sharp, computable in closed form, and yields the tight $Q_{\min}$ calibration.

6.2 Regime A: Entropy Accumulation Theorem

In Regime A, the m summands $p_C(x_i)$ share the single random variable C. They are not independent; they're conditionally i.i.d. given C.

A direct Chernoff bound *does not work*: the variance includes a between-circuit term that's not reduced by increasing m.

Aaronson and Hung handle this with the Entropy Accumulation Theorem (EAT) of Dupuis, Fawzi, and Renner (2020). EAT is a chain-rule-style decomposition that handles the conditional structure correctly. The high-level statement:

> Even when later rounds are conditioned on earlier rounds' outcomes, the total entropy accumulates linearly (modulo correction terms scaling with $\sqrt{m}$) provided per-round entropy is appropriately characterised.

For RCS with m shots on a single circuit, EAT gives a certified-entropy bound of the form

$$H_{\min}^{\mathrm{cert}} \geq m \cdot h(\phi) - O(\sqrt{m}) - \log(1/\varepsilon_{\mathrm{sou}}) \tag{15}$$

for some per-shot entropy rate $h(\phi)$.

6.3 Why Liu's bound is tighter

The crucial comparison: AH's bound has an $O(\sqrt{m})$ correction term; Liu's i.i.d. Chernoff has an $O(\log(1/\varepsilon)/\sqrt{m})$ correction in the deviation.

For typical parameters ($m \sim 10^3$, $\varepsilon_{\mathrm{sou}} \sim 10^{-6}$), the $O(\sqrt{m}) \approx 30$ correction in AH **eats into the certified-entropy budget** more than the $O(\log(1/\varepsilon))$ correction in Liu's i.i.d. regime.

This is why, for the same hardware and the same nominal F_{XEB}, Liu's structure yields *more* certified bits per round than AH's structure does. **Same physics; same expectation; tighter bound from the i.i.d. structure.**

7 Numerical comparison for Liu's parameter regime

For $n = 56$, $m = 1522$, $M = 30{,}010$, $\chi = 0.3$, $\varepsilon_{\mathrm{sou}} = 10^{-6}$, and the calibrated Liu adversary model ($4\times$ Frontier theoretical, eff $= 0.5$):

Quantity	Regime A (AH)	Regime B (Liu)
$\mathbb{E}[F_{\mathrm{XEB}}]$	ϕ (same)	ϕ (same)
$\mathrm{Var}[F_{\mathrm{XEB}}]$	$\sim c(\phi)/m$	$\sim c(\phi)/m$
Concentration tool	EAT	i.i.d. Chernoff
Soundness-eating correction	$O(\sqrt{m})$	$O(\log(1/\varepsilon))$
Certified bits per round	smaller	larger
$Q_{\min}$ (Liu's parameters)	$\approx$ larger†	1297
$H_{\min}^{\mathrm{cert}}$	smaller†	71,313 bits

† "Larger $Q_{\min}$" for AH means the threshold is more conservative — it certifies less for the same acceptance. Exact AH numbers depend on the EAT-derived bound; the qualitative point is that AH is strictly looser for the same parameters.

8 Implications for Quip

Three concrete implications for the protocol design:

1. Liu's structure is the right choice. For the same hardware and the same F_{XEB} score, Liu's i.i.d. structure certifies more bits per round than AH's. The v0.4 refactor adopted this structure, and this document explains *why* that adoption was the right call.

2. $\mathbf{F_{XEB}}$ computation is structure-agnostic. The formula (1) is identical in both regimes. The function in `leg5_certified_entropy.compute_fxeb` operates on (sample, probability) pairs without caring how the samples were grouped. This is why verifying our $\mathrm{F_{XEB}}$ implementation against Liu's experimental data is a meaningful test of the formula, independent of the structural choice.

3. The structural choice is separable from hardware. Whether we run on Quantinuum H2, IonQ Forte, or another QPU, the expected F_{XEB} is set by hardware fidelity. The structural choice (AH vs Liu) is independent: it's a protocol-level decision about how to allocate circuits to rounds. Hardware decisions and structural decisions can be made independently.

9 Summary

The two questions and their answers:

1. **Is the statistical distribution of $p_{C_i}(x_i)$ identical in the two regimes?**
   Yes. In both regimes, $p_{C_i}(x_i)$ follows the size-biased Porter-Thomas distribution, with mean $(1+\phi)/N$.
2. **Is the concentration behaviour of $\mathbf{F_{XEB}}$ identical in the two regimes?**
   No. Regime B's samples are independent, admitting direct Chernoff concentration. Regime A's samples are correlated through the shared circuit, requiring the Entropy Accumulation Theorem and producing a strictly looser certified-entropy bound.

The expected F_{XEB} is identical; the variance has the same leading $1/m$ scaling; but the *tail behaviour* differs in a way that propagates through the certified-entropy chain. Liu's structure is optimal among the two; this is what v0.4 implements.

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